S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026

RISK PULSE

From narrative noise to portfolio impact

Auditable financial risk signals from news and social text, connected directly to transparent wholesale-portfolio stress testing.

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College name to be added before submission

8 unique documents **9 risk signals** **4 stress runs**

Risk intelligence breaks at the handoff

THE GAP

- News and social posts arrive in incompatible formats and repeat the same event.
- A sentiment label alone does not describe event type, severity, confidence, or exposure.
- Manual handoffs make the path from narrative to portfolio impact slow and hard to audit.

THE RISK PULSE APPROACH

1	**Normalize**	News + social + replay
2	**Understand**	Entity + sentiment + event
3	**Score**	Impact + confidence + evidence
4	**Act**	Automatic, idempotent stress test

Design goal: one traceable path from source evidence to portfolio loss drivers

RISK PULSE

Auditable financial risk intelligence - source to portfolio impact



CONTROL PLANE

data lineage + hashes | model/config versions | confidence gates | structured logs + health | tests + CI + Docker

03 - AI/NLP RISK ENGINE

A signal is a contract, not a black-box label

Required outputs are bounded and augmented with confidence, evidence, provenance, and versions.

SENTIMENT

-1.0 to +1.0

P(positive) - P(negative)

EVENT

10-class taxonomy

explicit other / abstention

IMPACT

1.0 to 10.0

severity and urgency

EXPLAINABLE IMPACT FORMULA

Event severity	30%
Entity relevance	15%
Corroboration	15%
Tone | confidence | novelty | source	40%

ACTION GATE

- Event confidence >= 0.65
- Impact confidence >= 0.60
- Impact strictly > 7.0
- Supported scenario + idempotency

High-impact credit signal to explainable portfolio loss

Illustrative valuation across loans, bonds, and derivatives; every position reconciles.

PRE-STRESS VALUE	POST-STRESS VALUE	SCENARIO P&L	IMPACT TRIGGER
$411.8M	$404.4M	-$7.4M	8.9/10

CREDIT DETERIORATION

Illustrative issuer/sector credit event with spillover.

Equity: -16%

Rates: +25 bps

Credit spread: +300 bps

PD multiplier: 2.00x

VALUATION CHANNELS

LOANS	change in EAD x PD x LGD
BONDS	duration + convexity + spread
DERIVATIVES	delta / DV01 / FX sensitivity

05 - MEASURED RESULTS

Reproducible evidence, with limitations attached

The submitted sets verify contracts and behavior; they do not estimate production accuracy.

UNIQUE DOCUMENTS

8

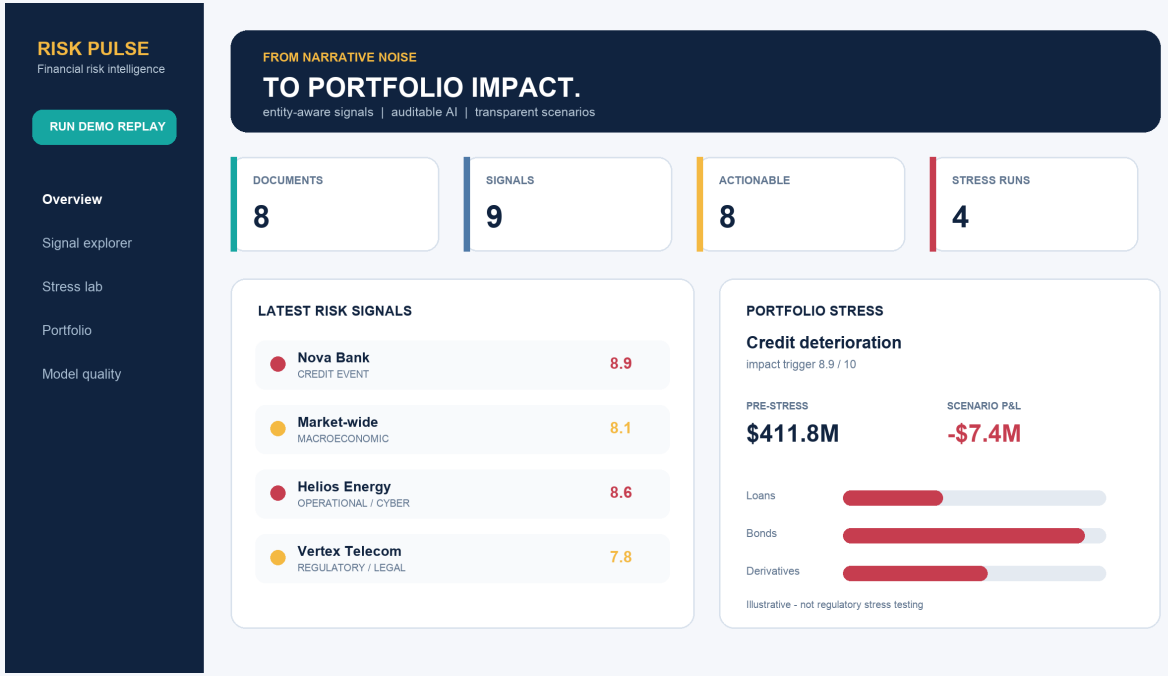
REPLAY FAILURES

0

SYNTHETIC-SET MODEL EVIDENCE

BASELINE	N	MACRO-F1
Finance sentiment	15	**100.00%**
Event taxonomy	20	**94.67%**

Compact synthetic data - not a production accuracy claim



Useful because it is honest about uncertainty

LIMITATIONS

- English-only, compact synthetic evaluation.

- Rules and lexicons miss novel phrasing and sarcasm.

- Impact contains transparent modeling judgment.

- Valuation is illustrative, not regulatory capital analysis.

NEXT STEPS

- Evaluate FinBERT and learned event models on larger time-split data.

- Calibrate impact against realized abnormal movement where licensing permits.

- Add analyst feedback, drift monitoring, and scenario sensitivity.

- Add tactical rebalancing only after the flagship risk path remains green.

RiskPulse: evidence -> signal -> scenario -> portfolio driver